

```

/* =====
HEADER
===== */

#include <LPC21XX.H>
#include <stdio.h>
typedef unsigned int u32;
typedef unsigned char u8;
typedef char s8;

/* delay */
void delay_ms(unsigned int ms);

/* spi drivers */
void spi0_init(void);
u8 spi0_data(u8);

/* ext adc drivers */
u32 mcp3204_read(u8);

/* uart0 drivers */
void uart0_init(u32);
void uart0_tx(u8);
u8 uart0_rx(void);
void uart0_tx_string(char *);
void uart0_tx_integer(u32);
void uart0_tx_float(float);

/* i2c drivers */
void i2c_init(void);
void i2c_send(u8 sa, u8 mr, u8 data);
u8 i2c_read(u8 sa, u8 mr);

/* lcd 4-bit driver */
void lcd_init(void);
void lcd_data(unsigned char data);

```

```
void lcd_cmd(unsigned char cmd);
```

```
void lcd_string(char *ptr);
```

```
void lcd_integer(int num);
```

```
void lcd_float(float f);
```

```
/* ADC DRIVERS */
```

```
void adc_init(void);
```

```
u32 adc_read(u8 ch);
```

```
/* =====
```

```
MAIN CODE
```

```
===== */
```

```
int main()
```

```
{
```

```
    u8 hr, min, sec, day, t;
```

```
    float vout1, tempr, vout2;
```

```
    unsigned long int adcout1, adcout2, light_perc;
```

```
    u32 ret;
```

```
    char arr[][4] = {"", "SUN", "MON", "TUE", "WED", "THU", "FRI", "SAT"};
```

```
    lcd_init();
```

```
    i2c_init();
```

```
    spi0_init();
```

```
    adc_init();
```

```
    uart0_init(9600);
```

```
    /* setting time */
```

```
    i2c_send(0xD0, 0x2, 0x11);
```

```
    i2c_send(0xD0, 0x1, 0x00);
```

```
i2c_send(0xD0, 0x0, 0x55);
```

```
i2c_send(0xD0, 0x3, 0x03);
```

```
uart0_tx_string("DATA ACQUISITION SYSTEM \n\r");
```

```
while (1)
```

```
{
```

```
    hr = i2c_read(0xD1, 0x2);
```

```
    min = i2c_read(0xD1, 0x1);
```

```
    sec = i2c_read(0xD1, 0x0);
```

```
    day = i2c_read(0xD1, 0x3);
```

```
    t = hr;
```

```
    hr = hr & 0x1f;
```

```
    lcd_cmd(0x80);
```

```
    lcd_data((hr / 0x10) + 48);
```

```
    lcd_data((hr % 0x10) + 48);
```

```
    lcd_data(':');
```

```
    lcd_data((min / 0x10) + 48);
```

```
    lcd_data((min % 0x10) + 48);
```

```
    lcd_data(':');
```

```
    lcd_data((sec / 0x10) + 48);
```

```
    lcd_data((sec % 0x10) + 48);
```

```
    lcd_data(':');
```

```
    uart0_tx_string("\r\n");
```

```
    uart0_tx((hr / 0x10) + 48);
```

```
    uart0_tx((hr % 0x10) + 48);
```

```
    uart0_tx(':');
```

```
    uart0_tx((min / 0x10) + 48);
```

```
uart0_tx((min % 0x10) + 48);  
uart0_tx(':');  
uart0_tx((sec / 0x10) + 48);  
uart0_tx((sec % 0x10) + 48);
```

```
if (((t >> 5) & 1) == 1)  
    uart0_tx_string("PM");  
else  
    uart0_tx_string("AM");
```

```
uart0_tx_string("\r\n");
```

```
if (((t >> 5) & 1) == 1)  
    lcd_string("PM");  
else  
    lcd_string("AM");
```

```
lcd_data(' ');  
lcd_string(arr[day]);  
uart0_tx_string(" ");  
uart0_tx_string(arr[day]);
```

```
/* Temperature Sensor LM35 */  
adcout1 = adc_read(1);  
vout1 = (adcout1 * 3.3f) / 1023.0f;  
tempr = (vout1 - 0.5f) / 0.01f;
```

```
uart0_tx_string("T:");  
uart0_tx_float(tempr);  
uart0_tx_string(" C");
```

```

    lcd_cmd(0xc0);

    lcd_string("T");

    lcd_float(tempr);


    /* POT Reading */
    uart0_tx_string(" pot:");
    adcout2 = adc_read(2);
    vout2 = (adcout2 * 3.3f) / 1023.0f;


    uart0_tx_float(vout2);
    uart0_tx_string("V ");


    lcd_string(" P");
    lcd_float(vout2);
    lcd_data('V');


    /* LDR (MCP3204 via SPI) */
    ret = mcp3204_read(2);
    light_perc = 100 - ((ret * 100) / 4095);


    uart0_tx_string(" Light:");
    uart0_tx_integer(light_perc);
    uart0_tx_string("%\r\n");


    lcd_data(' ');
    lcd_string("L");
    lcd_integer(light_perc);
    lcd_data('%');
}

return 0;

```

```
}
```

```
/* =====
```

```
I2C DRIVER
```

```
===== */
```

```
#define SI ((I2CONSET >> 3) & 1)
```

```
void i2c_init(void)
```

```
{
```

```
    PINSEL0 |= 0x50;
```

```
    I2SCLH = 75;
```

```
    I2SCLL = 75;
```

```
    I2CONSET = (1 << 6);
```

```
}
```

```
void i2c_send(u8 sa, u8 mr, u8 data)
```

```
{
```

```
    I2CONSET = (1 << 5);
```

```
    I2CONCLR = (1 << 3);
```

```
    while (!SI)
```

```
        ;
```

```
    I2CONCLR = (1 << 5);
```

```
    I2DAT = sa;
```

```
    I2CONCLR = (1 << 3);
```

```
    while (!SI)
```

```
        ;
```

```
    I2DAT = mr;
```

```
    I2CONCLR = (1 << 3);
```

```
    while (!SI)
```

;

I2DAT = data;

I2CONCLR = (1 << 3);

while (!SI)

;

I2CONSET = (1 << 4);

I2CONCLR = (1 << 3);

}

u8 i2c_read(u8 sa, u8 mr)

{

u8 data;

I2CONSET = (1 << 5);

I2CONCLR = (1 << 3);

while (!SI)

;

I2CONCLR = (1 << 5);

I2DAT = (sa & 0xFE);

I2CONCLR = (1 << 3);

while (!SI)

;

I2DAT = mr;

I2CONCLR = (1 << 3);

while (!SI)

;

```

I2CONSET = (1 << 5);
I2CONCLR = (1 << 3);
while (!SI)
    ;
I2CONCLR = (1 << 5);

I2DAT = sa;
I2CONCLR = (1 << 3);
while (!SI)
    ;

I2CONCLR = (1 << 3);
while (!SI)
    ;
data = I2DAT;

I2CONSET = (1 << 4);
I2CONCLR = (1 << 3);

return data;
}

/* =====
SPI DRIVER
===== */

#define SPIF ((SOSPSR >> 7) & 1)

void spi0_init(void)
{
    PINSEL0 |= 0x1500;
    IODIRO |= (1 << 7);

```

```

IOSET0 = (1 << 7);
SOSPCR = 0x20;
SOSPCCR = 150;
}

```

```

u8 spi0_data(u8 data)
{
    SOSPDR = data;
    while (!SPIF)
        ;
    return SOSPDR;
}

```

```

/* =====
MCP3204 ADC SPI DRIVER
===== */

```

```

u32 mcp3204_read(u8 ch_num)
{
    u8 byteH, byteL;
    u32 result = 0;

    ch_num <= 6;

    IOCLR0 = (1 << 7);
    spi0_data(0x06);
    byteH = spi0_data(ch_num);
    byteL = spi0_data(0x00);
    IOSET0 = (1 << 7);

    byteH &= 0x0F;
    result = (byteH << 8) | byteL;
}

```

```

    return result;
}

/* =====
UART DRIVER
===== */

#define THRE ((UOLSR >> 5) & 1)
#define RDR (UOLSR & 1)

void uart0_init(u32 baud)
{
    unsigned int result = 0, pclk;
    int a[] = {15, 40, 30, 15, 15};
    pclk = a[VPBDIV] * 1000000;
    result = pclk / (16 * baud);

    PINSELO |= 5;
    UOLCR = 0x83;
    UODLM = (result >> 8) & 0xff;
    UODLL = result & 0xff;
    UOLCR = 3;
}

void uart0_tx(u8 data)
{
    UOTHR = data;
    while (!THRE)
        ;
}

```

```

u8 uart0_rx(void)
{
    while (!RDR)
        ;
    return UORBR;
}

```

```

void uart0_tx_string(char *p)
{
    while (*p)
    {
        uart0_tx(*p++);
    }
}

```

```

void uart0_tx_integer(u32 temp)
{
    char arr[10];
    sprintf(arr, "%d", temp);
    uart0_tx_string(arr);
}

```

```

void uart0_tx_float(float temp)
{
    char arr[16];
    sprintf(arr, "%.2f", temp);
    uart0_tx_string(arr);
}

```

```

/* =====

```

```

LCD DRIVER

```

```

===== */

void lcd_init()
{
    IODIR1 = (0xfe << 16);
    PINSEL2 = 0X0;
    lcd_cmd(0x02);
    lcd_cmd(0x28);
    lcd_cmd(0x0e);
    lcd_cmd(0x01);
}

void lcd_data(unsigned char data)
{
    IOCLR1 = (0xfe << 16);
    IOSET1 = ((data & 0xF0) << 16);

    IOSET1 = (1 << 17);
    IOCLR1 = (1 << 18);
    IOSET1 = (1 << 19);
    delay_ms(2);
    IOCLR1 = (1 << 19);

    IOCLR1 = (0xfe << 16);
    IOSET1 = ((data & 0x0F) << 20);

    IOSET1 = (1 << 17);
    IOCLR1 = (1 << 18);
    IOSET1 = (1 << 19);
    delay_ms(2);
    IOCLR1 = (1 << 19);
}

```

```
void lcd_cmd(unsigned char cmd)
```

```
{  
    IOCLR1 = (0xfe << 16);  
    IOSET1 = ((cmd & 0xf0) << 16);  
    IOCLR1 = (1 << 17);  
    IOCLR1 = (1 << 18);  
    IOSET1 = (1 << 19);  
    delay_ms(2);  
    IOCLR1 = (1 << 19);  
  
    IOCLR1 = (0xfe << 16);  
    IOSET1 = ((cmd & 0x0f) << 20);  
    IOCLR1 = (1 << 17);  
    IOCLR1 = (1 << 18);  
    IOSET1 = (1 << 19);  
    delay_ms(2);  
    IOCLR1 = (1 << 19);  
}
```

```
void lcd_string(char *ptr)
```

```
{  
    while (*ptr)  
        lcd_data(*ptr++);  
}
```

```
void lcd_integer(int num)
```

```
{  
    int i;  
    char a[10];
```

```
if (num == 0)
```

```
{
```

```
    lcd_data('0');
```

```
    return;
```

```
}
```

```
for (i = 0; num; i++, num /= 10)
```

```
    a[i] = num % 10 + '0';
```

```
while (--i >= 0)
```

```
    lcd_data(a[i]);
```

```
}
```

```
void lcd_float(float f)
```

```
{
```

```
    int num;
```

```
    if (f < 0)
```

```
    {
```

```
        lcd_data('-');
```

```
        f = -f;
```

```
    }
```

```
    num = (int)f;
```

```
    lcd_integer(num);
```

```
    lcd_data('.');
```

```
    num = (int)((f - num) * 100);
```

```
    lcd_integer(num);
```

```
}
```

```

/* =====
ADC DRIVER
===== */

void adc_init(void)
{
    PINSEL1 |= 0x15400000;
    ADCR = 0x00200400;
}

u32 adc_read(u8 ch_num)
{
    u32 result = 0;

    ADCR |= (1 << ch_num);
    ADCR |= (1 << 24);

    while (((ADDR >> 31) & 1) == 0)
        ;

    ADCR &= ~(1 << 24);
    ADCR &= ~(1 << ch_num);

    result = ((ADDR >> 6) & 0x3FF);

    return result;
}

/* =====
DELAY
===== */

void delay_ms(unsigned int ms)

```

```
{  
    TOPC = 0;  
    TOPR = 15000 - 1;  
    TOTC = 0;  
    TOTCR = 1;  
  
    while (TOTC < ms)  
        ;  
  
    TOTCR = 0;  
}
```